WORKED SOLUTION

STEP III 1997, Question 8

Pure • Diagonalising Quadratic Forms

STEP III 1997, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q8

The key idea

A symmetric matrix describes a quadratic form $ax^2 + 2bxy + cy^2$, and the conjugated matrix $R_{-\alpha}AR_\alpha$ describes the same form in axes rotated through α. Making it diagonal removes the xy term, so in the new axes the ellipse is in standard position and its semi-axes can be read off. Since rotations preserve distance from the origin, those semi-axes are the extreme distances we want.

Solution

Throughout, write

$$R_\alpha = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}, \qquad R_{-\alpha} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix} = R_\alpha^{\mathsf{T}}.$$

Part (i)

Let $C = \cos\alpha$ and $S = \sin\alpha$. First

$$AR_\alpha = \begin{pmatrix} aC + bS & -aS + bC \\ bC + cS & -bS + cC \end{pmatrix},$$

and multiplying on the left by $R_{-\alpha}$, the top-right entry is

$$\begin{aligned} C(-aS + bC) + S(-bS + cC) &= b(C^2 - S^2) - (a - c)SC \\ &= b\cos 2\alpha - \tfrac{1}{2}(a - c)\sin 2\alpha. \end{aligned}$$

Because A is symmetric, so is $R_{-\alpha}AR_\alpha = R_\alpha^{\mathsf{T}}AR_\alpha$ (its transpose is itself), so the bottom-left entry is the same and one condition does both. The matrix is diagonal exactly when

$$\tan 2\alpha = \frac{2b}{a - c} \quad (a \neq c), \qquad \text{so we may take} \qquad \alpha = \tfrac{1}{2}\arctan\frac{2b}{a - c}.$$

If $a = c$ the condition is $b\cos 2\alpha = 0$, and $\alpha = \frac{1}{4}\pi$ works (any α works if also $b = 0$, when A is already diagonal).

Part (ii)

Expanding the equation of the ellipse,

$$x^2 + y^2 + 4xy\cot 2\theta + 4x^2\cot^2 2\theta = 1,$$

while $\begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + cy^2$. Matching coefficients,

$$a = 1 + 4\cot^2 2\theta, \qquad b = 2\cot 2\theta, \qquad c = 1.$$

Then $a - c = 4\cot^2 2\theta \neq 0$ (as $0 < 2\theta < \frac{1}{2}\pi$), and

$$\frac{2b}{a-c} = \frac{4\cot 2\theta}{4\cot^2 2\theta} = \tan 2\theta,$$

so $\alpha = \theta$ satisfies the condition from part (i): $R_{-\theta}AR_\theta$ is diagonal.

For the diagonal entries, the same multiplication as in part (i) gives

$$\text{top-left} = aC^2 + 2bSC + cS^2, \qquad \text{bottom-right} = aS^2 - 2bSC + cC^2,$$

now with $C = \cos\theta$, $S = \sin\theta$. Put $k = \cot 2\theta$ and $\mu = \cos 2\theta$, and use $2SC = \sin 2\theta$, $k\sin 2\theta = \mu$, $C^2 = \frac{1}{2}(1+\mu)$, $S^2 = \frac{1}{2}(1-\mu)$, and $k^2 = \mu^2/(1-\mu^2)$. For the bottom-right entry,

$$\begin{aligned}(1+4k^2)S^2 - 2k\sin 2\theta + C^2 &= 1 + 4k^2S^2 - 2\mu \\ &= 1 - 2\mu + \frac{2\mu^2(1-\mu)}{(1-\mu)(1+\mu)} = \frac{(1-2\mu)(1+\mu)+2\mu^2}{1+\mu} \\ &= \frac{1-\mu}{1+\mu} = \frac{2\sin^2\theta}{2\cos^2\theta} = \tan^2\theta.\end{aligned}$$

In exactly the same way the top-left entry is $1 + 2\mu + \dfrac{2\mu^2}{1-\mu} = \dfrac{1+\mu}{1-\mu} = \cot^2\theta$. So

$$R_{-\theta}AR_\theta = \begin{pmatrix} \cot^2\theta & 0 \\ 0 & \tan^2\theta \end{pmatrix}.$$

Ask yourself Is there a quick way to confirm these two entries without redoing the algebra?

Yes. Conjugating doesn't change the trace or the determinant. Here $\det A = ac - b^2 = 1 + 4k^2 - 4k^2 = 1$, and indeed $\cot^2\theta\tan^2\theta = 1$. Also $\operatorname{tr} A = 2 + 4\cot^2 2\theta$, and since $\cot\theta - \tan\theta = \dfrac{\cos^2\theta - \sin^2\theta}{\sin\theta\cos\theta} = 2\cot 2\theta$,

$$\cot^2\theta + \tan^2\theta = (\cot\theta - \tan\theta)^2 + 2 = 4\cot^2 2\theta + 2.$$

Part (iii)

Introduce new coordinates by $\begin{pmatrix} x \\ y \end{pmatrix} = R_\theta \begin{pmatrix} X \\ Y \end{pmatrix}$, so that the X- and Y-axes are the old axes rotated through θ. Since $\begin{pmatrix} x & y \end{pmatrix} = \begin{pmatrix} X & Y \end{pmatrix} R_\theta^{\mathsf{T}} = \begin{pmatrix} X & Y \end{pmatrix} R_{-\theta}$, the equation of the ellipse becomes

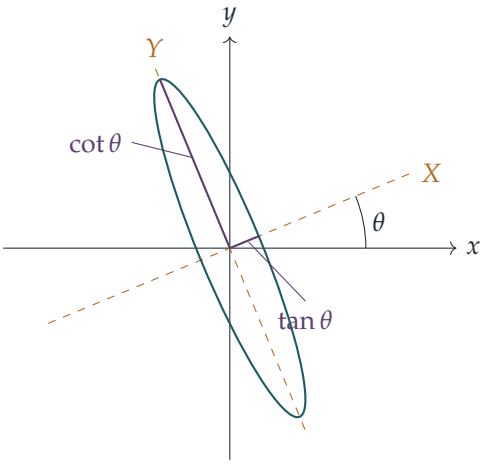
$$\begin{pmatrix} X & Y \end{pmatrix} R_{-\theta} A R_\theta \begin{pmatrix} X \\ Y \end{pmatrix} = 1, \qquad \text{that is} \qquad X^2 \cot^2\theta + Y^2 \tan^2\theta = 1.$$

This is symmetric under $(X, Y) \mapsto (-X, -Y)$, so the centre is the origin. A rotation preserves lengths, so the distance from the centre to a point is $r = \sqrt{x^2 + y^2} = \sqrt{X^2 + Y^2}$.

Because $0 < \theta < \frac{1}{4}\pi$, we have $0 < \tan\theta < 1 < \cot\theta$, so $\tan^2\theta < \cot^2\theta$. For any point on the ellipse,

$$\tan^2\theta\,(X^2 + Y^2) \;\le\; X^2\cot^2\theta + Y^2\tan^2\theta = 1 \;\le\; \cot^2\theta\,(X^2 + Y^2),$$

so $\tan^2\theta \le r^2 \le \cot^2\theta$, that is, $\tan\theta \le r \le \cot\theta$. The left inequality is an equality when $X = 0$, at $(X, Y) = (0, \pm\cot\theta)$, and the right one when $Y = 0$, at $(X, Y) = (\pm\tan\theta, 0)$. Both bounds are attained, so the minimum and maximum distances from the centre to the ellipse are $\tan\theta$ and $\cot\theta$, as required.



The figure shows the case $\theta = \frac{1}{8}\pi$, with the rotated axes dashed.

Checks

- $(0, \pm 1)$ lies on the ellipse, and $\tan\theta < 1 < \cot\theta$, consistent with the bounds.
- For $\theta = \frac{1}{8}\pi$ the major-axis end in the original coordinates is $\cot\theta\,(-\sin\theta, \cos\theta) = (-\cos\theta, \cos^2\theta/\sin\theta) \approx (-0.924,\, 2.230)$. Here $\cot 2\theta = 1$, and $0.924^2 + (2.230 - 1.848)^2 \approx 0.854 + 0.146 = 1$, as it should be.

- Numerically, scanning 200 000 directions round the ellipse for several values of θ gave, for $\theta = 0.3$, a minimum distance of 0.309336 and a maximum of 3.232728, matching $\tan 0.3$ and $\cot 0.3$.

Common slips

- Matching the xy coefficient to b rather than $2b$: the form is $ax^2 + 2bxy + cy^2$.
- Mixing up which rotation goes where: with $\mathbf{x} = R_\theta \mathbf{X}$ the row vector is $\mathbf{X}^{\mathsf{T}} R_{-\theta}$, which is why $R_{-\theta} A R_\theta$ appears.
- Thinking the axis with the larger coefficient is the longer one: $\cot^2\theta X^2$ has the larger coefficient, so the ellipse is *short* in the X direction.

Beyond STEP

This is the two-dimensional case of the spectral theorem: every real symmetric matrix can be diagonalised by a rotation, and its eigenvalues are the diagonal entries. The eigenvalues of a quadratic form classify conics and quadric surfaces, and the same idea gives principal axes of inertia in mechanics and principal components in statistics. Strang's *Introduction to Linear Algebra* treats symmetric matrices and quadratic forms very readably.

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